Adaptive Handover Detection for QUIC over LEO Satellite Networks

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***Abstract*—Low Earth Orbit (LEO) satellites travel at approximately 7.5 km/s, causing frequent link handovers that manifest as round-trip time (RTT) spikes and transient packet loss events. Traditional transport-layer congestion control algorithms misinterpret these link-level disturbances as network congestion, leading to unnecessary reductions in the congestion window (*cwnd*) and consequent throughput degradation. Existing optimization schemes either depend on proprietary satellite telemetry that is unavailable to end-users, or exploit fixed handover periodicity patterns whose applicability is restricted to specific constellations and geographic regions. This paper proposes the Adaptive Handover Detection (AHD) algorithm, which operates exclusively on observable end-to-end metrics, specifically RTT, available bandwidth, and packet loss rate, to distinguish link handover events from genuine network congestion without requiring any external auxiliary information. Integrated as a transparent wrapper around BBRv3 within the QUIC protocol stack and evaluated in a Mininet-based simulation environment driven by real Starlink measurement traces, AHD identifies handover-induced network state transitions, suppresses unnecessary *cwnd* reductions, increases QUIC throughput by up to 32%, and reduces the packet loss rate by 11% relative to standard BBR. An ablation study confirms that both EWMA-based smoothing and the freeze protection mechanism are essential contributors to the observed performance gains. A sensitivity analysis of all five tunable parameters justifies the selected configuration and demonstrates that performance remains robust throughout the evaluated parameter range.**

***Index Terms*—QUIC, congestion control, LEO satellite networks, handover detection**

I. Introduction

Satellite communication has emerged as a key enabler of global connectivity, with Low Earth Orbit (LEO) satellite networks representing one of the most rapidly advancing paradigms in this domain. Compared with geostationary (GEO) satellites and terrestrial infrastructure, LEO constellations offer broad geographic coverage, terrain independence, high link reliability, reduced propagation latency, and lower susceptibility to physical obstructions [10]. Starlink, operated by SpaceX, exemplifies the commercial maturity of this technology, with more than 9,000 satellites in operation as of January 2026, serving approximately nine million subscribers across nearly 155 countries and regions [11].

A key challenge faced by LEO satellite networks stems from the dynamic changes caused by satellite mobility. Each satellite must orbit the Earth at approximately 7.5 km/s to resist gravity and maintain its orbit. This rapid movement, together with the associated handovers and dynamic topology changes, adversely affects the stability and reliability of the network [6]. Each handover event introduces delay spikes and packet loss, which cause the congestion control (CC) algorithm to perceive network congestion and proactively reduce the congestion window (*cwnd*). During recovery, the algorithm enters a slow-start state, which has a significant negative impact on latency-sensitive applications such as low-latency live video streaming [12]. Among CC algorithms, Bottleneck Bandwidth and Round-trip propagation time (BBR) has been shown to achieve higher throughput and better bandwidth fairness than CUBIC in satellite network environments [4], [5], making it a strong baseline for further improvement.

Several studies have sought to optimize communication protocol performance by predicting handover events. SaTCP [6] attempts to incorporate satellite location information, but the link-layer feedback it relies upon is difficult to obtain in commercial, closed-source systems. Although StarQUIC [7] exploits specific handover periodicity patterns, its generalizability is limited by regional and constellation-specific variations. This paper proposes an adaptive algorithm that relies solely on end-to-end measurement data without requiring external auxiliary information. The principal contributions of this paper are: (1) an AHD detection mechanism based on the BBRv3

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algorithm that is suitable for the QUIC protocol; and (2) a reproducible simulation environment built on real Starlink datasets to verify the algorithm's advantages in improving throughput and robustness.

II. Related Work

Due to the characteristics of LEO satellites, such as high-speed movement, frequent link changes, and significant path delay fluctuations, traditional Internet transport protocols often struggle to maintain stable performance in this environment. Consequently, considerable research effort has been directed toward the performance of transport protocols in LEO satellite network environments.

Khan et al. [2] evaluated the performance of various transport protocols in LEO network environments through simulation and experimentation, demonstrating that path changes and delay fluctuations caused by satellite orbital movement significantly impair the stability of CC mechanisms. Barbosa et al. [3] systematically compared the performance of TCP CC algorithms in LEO networks, concluding that traditional CC algorithms often fail to adapt to frequent link changes, leading to decreased throughput and performance variance.

Cao and Zhang [6] proposed SaTCP, which is based on CUBIC, to adjust TCP CC decisions by predicting satellite handover events or route update timestamps using the predictability of satellite positions. SaTCP requires ground stations (GS) to report interruption events in advance, enabling user terminals (UT) to notify user equipment (UE) of upcoming handover events. However, these mechanisms do not exist in commercial LEO satellite networks, where such devices are inaccessible to third-party developers. Consequently, the evaluation of SaTCP is limited to the LeoEM simulator and does not extend to real-world deployment.

QUIC is a UDP-based transport layer protocol standardized by the IETF (RFC 9000) [9], featuring encryption, multiplexing, and low latency. It combines the reliability of TCP with the flexibility of UDP, shortens connection establishment time through 0-RTT and 1-RTT handshakes, and supports connection migration based on Connection Identifier (CID) identification, enabling seamless adaptation to network handover scenarios. Its CC module supports multiple algorithms, including CUBIC and BBR, and its user-space implementation makes deployment and iteration more flexible.

Wang et al. [4] evaluated the performance of QUIC with BBR in satellite Internet scenarios using a dedicated simulation platform and found that QUIC+BBR outperforms QUIC+CUBIC in terms of throughput and goodput under typical packet loss rates of 1–10% in satellite networks, while maintaining a stable transmission rate and good bandwidth fairness. Bujari et al. [5] further demonstrated that BBR is more efficient than alternative CC algorithms within QUIC in most scenarios. Both studies indicate that targeted improvements to BBR are necessary to further enhance its performance in satellite networks.

Kamel et al. [7] observed that Starlink handover events occur at approximately 12, 27, 42, and 57 seconds per minute [8], leading to congestion misjudgment by QUIC's CC algorithm. They consequently proposed StarQUIC, a *cwnd* freeze mechanism that is independent of the CC algorithm. This mechanism exploits predictable handover patterns to avoid unnecessary *cwnd* reductions, reducing QUIC transmission completion times by up to 35%. However, Starlink experiences slight timing variations across different regions, and other LEO satellite constellations do not exhibit this pattern, limiting StarQUIC's generalizability.

This paper proposes the Adaptive Handover Detection (AHD) algorithm, evaluated using a Mininet-based emulator and a publicly available Starlink latency dataset [1] for reproducibility. In this work, comparison with SaTCP is precluded because SaTCP requires proprietary link-layer feedback that is unavailable in the simulation environment. Comparison with machine learning-based approaches is similarly deferred, as such methods require labeled handover datasets and introduce non-deterministic inference latency. Therefore, the present evaluation focuses on demonstrating handover detection performance against BBR and StarQUIC.

III. Methodology

In LEO networks, handover events cause short-term network state transitions that are typically manifested as an increase in RTT and a decrease in available bandwidth. When this phenomenon is misclassified as regular congestion, the CC algorithm triggers unnecessary *cwnd* reduction, resulting in a significant decrease in throughput and an extended recovery time.

To address this, AHD is designed as an adaptive handover detection mechanism that operates as an outer wrapper around the base CC algorithm (BBRv3):

- During stable link periods, the base CC algorithm is used for rate control.
- Upon identification of a handover signature rather than a congestion signature, AHD enters a freeze protection phase to suppress *cwnd*-reducing signals.
- AHD restores the learning baseline gradually after the freeze concludes.

The primary objective of this mechanism is to maintain the stability of the transmission window during handover disturbances and reduce the probability of erroneous *cwnd* reduction.

A. Handover Event Detection

Upon each ACK reception, AHD extracts three observable transport metrics:

$$\mathbf{x}_t = (\mathrm{rtt}_t,\ \mathrm{bw}_t,\ \mathrm{loss}_t) \tag{1}$$

A per-metric statistical baseline is maintained using an Exponential Weighted Moving Average (EWMA):

$$\bar{\mathbf{x}}_t = (1-\alpha)\,\bar{\mathbf{x}}_{t-1} + \alpha\,\mathbf{x}_t \tag{2}$$

RTT and bandwidth use a faster update coefficient ($\alpha = 0.15$), while the packet loss rate uses a slower update coefficient ($\alpha = 0.05$), in order to suppress the effect of

instantaneous jitter on the loss baseline. The relative deviations for RTT and bandwidth are then computed as:

$$e_t^{\text{rtt}} = \frac{\text{rtt}_t - \overline{\text{rtt}}_t}{\overline{\text{rtt}}_t}, \qquad e_t^{\text{bw}} = \frac{\overline{\text{bw}}_t - \text{bw}_t}{\overline{\text{bw}}_t} \tag{3}$$

A handover-type event is flagged when all three of the following conditions are satisfied simultaneously:

$$e_t^{\text{rtt}} > \theta_{\text{rtt}}, \quad e_t^{\text{bw}} > \theta_{\text{bw}}, \quad \text{loss}_t < \theta_{\text{loss}} \tag{4}$$

This joint condition captures the combined signature of a sudden increase in latency, a sudden decrease in bandwidth, and the absence of elevated packet loss, and is used to distinguish between link handover and queue congestion.

B. Prediction and Error Control Mechanism

AHD accumulates evidence of handover-type conditions using an event counter C_{event} and defines the prediction error as:

$$E_t = \frac{|e_t^{\text{rtt}}| + |e_t^{\text{bw}}|}{2} \tag{5}$$

The metric E_t averages the normalized deviations of RTT and bandwidth. Because both e_t^{rtt} and e_t^{bw} are dimensionless relative deviations, obtained by normalizing the raw measurement differences by their respective EWMA baseline values, this equal-weighted average provides a unit-free, combined error signal that penalizes large deviations in either metric without requiring explicit scaling factors.

The conditions for entering the freeze state, in which the predicted handover is confirmed, are:

$$C_{\text{event}} \geq N_{\text{th}} \ \wedge \ E_t \leq E_{\max} \tag{6}$$

If the event threshold has been reached but the error remains persistently elevated, the cumulative consecutive error counter C_{err} is incremented. A Refine operation is triggered when either of two conditions holds: (i) consecutive prediction errors persist ($C_{\text{err}} \geq 5$), indicating a stale baseline; or (ii) no ACK is received for 2 seconds (timeout), implying a possible route change or silent loss. These two conditions are independent and operate under OR logic: whichever occurs first triggers a reset of the event and error counters, and the RTT and bandwidth baselines are rebuilt from the current link estimate to prevent a stale model from amplifying detection errors.

C. Freeze Protection Strategy

After predicting the handover, the algorithm enters a freeze period T_f:

$$t_{\text{freeze_until}} = t_0 + t_f \tag{7}$$

The freeze duration is RTT-adaptive: the soft window is set to $T_{\text{soft}} = \text{clip}(2 \cdot \text{SRTT},\ 250\,\text{ms},\ 700\,\text{ms})$, which covers the typical LEO handover transient. After T_{soft} expires, the freeze is released only when two consecutive ACK samples produce no further anomaly, confirming a stable path. A hard cap $T_{\text{hard}} = 2T_{\text{soft}}$ bounds the worst case, and the freeze is forcibly released whenever the single-sample loss rate or the EWMA loss rate exceeds the threshold, ensuring that AHD does not hold the window high during genuine congestion.

During the freeze period, two key controls are implemented:

- **Window floor enforcement:** The *cwnd* value at freeze entry ($cwnd_f$) is recorded, and $cwnd(t) \geq cwnd_f$ is enforced to prevent *cwnd* from dropping due to handover interference.
- **Notification suppression:** Negative congestion notifications that would reduce *cwnd*, such as repeated packet loss signals and timeouts, are masked. ACK-driven internal state updates, however, continue normally, keeping bandwidth estimates and counters current.

After the freeze concludes, if the observation is stable (the event count has been reset to zero and remains so for several cycles), the freeze exits, the baseline is refreshed with the latest smoothed RTT and bandwidth values, and the accumulated loss history is gradually decayed, completing the parameter update phase.

D. Parameter Sensitivity Analysis

To assess the robustness of the AHD scheme and justify the selected parameter configuration, a sensitivity analysis was conducted on five key parameters: θ_{rtt}, θ_{bw}, θ_{loss}, N_{th}, and $E_{\max}$. Fig. 1 illustrates the impact of these parameters on average network throughput.

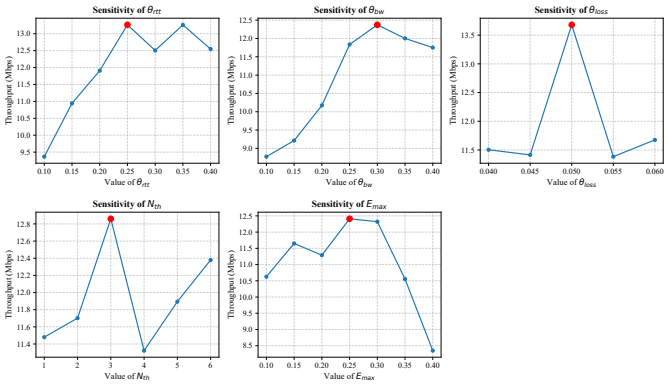


Fig. 1. Parameter sensitivity analysis of the proposed AHD scheme. Subplots (a)–(e) show the impact of different parameter values on average throughput. The red markers indicate the optimal parameter settings selected for the finalized algorithm: (a) $\theta_{rtt} = 0.25$, (b) $\theta_{bw} = 0.3$, (c) $\theta_{loss} = 0.05$, (d) $N_{th} = 3$, and (e) $E_{max} = 0.25$.

The key observations from each parameters are as follows:

- **Threshold Parameters ($\theta_{\text{rtt}}, \theta_{\text{bw}}, \theta_{\text{loss}}$):** As shown in Fig. 1(a)–(c), throughput initially increases as the thresholds are tuned to capture handover characteristics more accurately. However, performance degrades when the thresholds exceed their optimal values ($\theta_{\text{rtt}} = 0.25$, $\theta_{\text{bw}} = 0.3$, and $\theta_{\text{loss}} = 0.05$). Excessively high thresholds may cause missed detections of handover events, while overly low thresholds may trigger false positives under normal channel jitter.
- **Observation Window (N_{th}):** Fig. 1(d) indicates that $N_{\text{th}} = 3$ yields the best result. A smaller N_{th} renders the algorithm overly sensitive to transient noise, whereas

a larger N_{th} introduces detection delay that hinders timely window adjustment during handovers.

- **Maximum Estimation Error ($E_{\max}$):** The sensitivity of $E_{\max}$ is shown in Fig. 1(e). The optimal value is 0.25. When $E_{\max}$ increases beyond 0.3, throughput drops to approximately 8.35 Mbps, suggesting that an over-tolerant error margin prevents the algorithm from accurately distinguishing between congestion and link state changes.

Based on these results, the optimal parameter set $\{\theta_{\text{rtt}} = 0.25,\ \theta_{\text{bw}} = 0.3,\ \theta_{\text{loss}} = 0.05,\ N_{\text{th}} = 3,\ E_{\max} = 0.25\}$ is adopted for all subsequent performance evaluations.

IV. Evaluation & Results

AHD was evaluated against standard BBR (BBRv3) and StarQUIC in a controlled simulation environment driven by real Starlink measurement traces. Four key performance indicators were assessed: throughput, flow completion time, latency, and packet loss rate.

A. Evaluation

The simulation platform is based on Mininet and utilizes measurement data collected in [1]. These traces include IRTT delay measurements collected at 10 ms intervals and iPerf3 throughput measurements collected at 100 ms intervals. The bandwidth and delay of the links are updated every 100 ms based on the traces using the `tc-netem` and `tc-tbf` utilities available on Linux. This configuration results in a probed packet loss rate of approximately 2–4% in both directions. Although packet loss was not altered during handover events, the peak RTTs captured through tracing cause picoquic to perceive packet loss via retransmission timeout (RTO). The evaluation uses only non-ISL Starlink traces from the Seattle PoP [1]. ISL traces are excluded due to dataset limitations and to isolate satellite-to-ground handover effects; accordingly, all claims are scoped to non-ISL LEO access links, and extension to ISL-enabled topologies is left as future work.

For each test, a 256 MB file was transferred from the server to the client using BBR, StarQUIC, and AHD, respectively, and the transfer completion time was recorded. Ten runs were performed per trace over four traces. The simulated network topology is shown in Fig. 2.

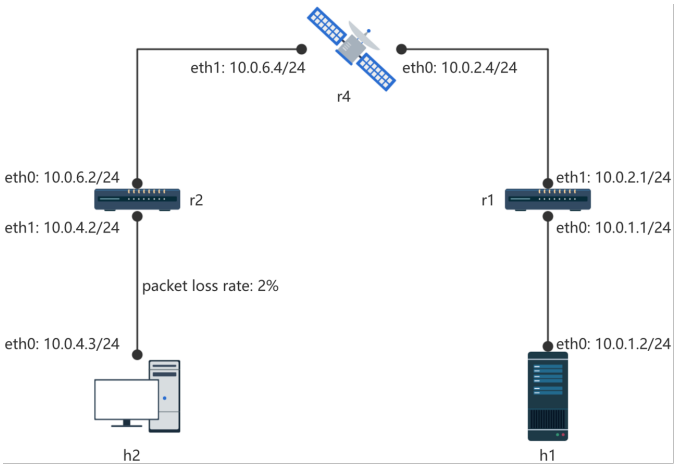


Fig. 2. Network topology diagram of simulated environment.

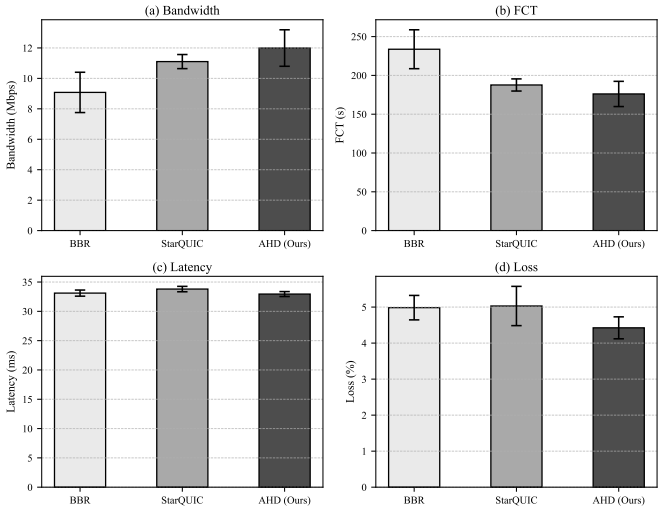


Fig. 3. Performance comparison under dynamic LEO scenarios. The error bars represent the 95% confidence intervals ($n = 10$).

B. Results

The effectiveness of the proposed AHD optimization scheme is verified by comparison with standard QUIC (based on BBRv3) and StarQUIC. Fig. 3 presents the statistical results of four key performance indicators, where the bars represent mean values and the error bars indicate 95% confidence intervals (CI) based on $n = 10$ independent test runs.

As shown in Fig. 3(a), the average throughput of AHD reached 12 Mbps, representing an improvement of approximately 32.11% over standard BBR (9.08 Mbps) and 9.8% over StarQUIC (11.1 Mbps). Although AHD exhibits a slightly larger standard deviation than StarQUIC, its superior mean throughput demonstrates a greater ability to aggregate bandwidth in the highly dynamic LEO satellite environment.

The FCT results are presented in Fig. 3(b). AHD reduces the average FCT to 176.10 s, compared with 233.72 s for BBR and 187.67 s for StarQUIC, representing reductions of 24.65% and 6.16%, respectively. The error bars for BBR are substantially wider than those of AHD and StarQUIC, indicating that conventional BBR experiences high performance variance during satellite handovers. By contrast, AHD maintains a tighter confidence interval, demonstrating its effectiveness in providing stable transmission performance over LEO satellite networks.

Due to frequent handovers and channel interference, LEO satellite networks often exhibit high jitter and non-congestion packet loss. As shown in Fig. 3(c), the RTT of all schemes remains approximately 33 ms. AHD achieves the lowest average latency at 32.95 ms, marginally below BBR (33.11 ms) and StarQUIC (33.60 ms). The small magnitude of these

differences confirms that AHD improves throughput without inducing additional queuing delay.

As shown in Fig. 3(d), the average packet loss rate of AHD is reduced to 4.42%, representing relative reductions of 11.04% and 11.92% compared with BBR (4.98%) and StarQUIC (5.03%), respectively. The upper bound of AHD's 95% CI remains below the mean values of both BBR and StarQUIC. This confirms that the proposed threshold-based detection mechanism can accurately distinguish between congestion-related losses and non-congestion-related losses, such as handover-induced drops, thereby preventing the window collapse effect and reducing redundant retransmissions. This distinction is important for latency-sensitive applications such as live video streaming, as it lowers the probability of frame freezes.

C. Ablation Study

To quantify the contribution of each removable component of AHD, two variants were evaluated: (i) *No-EWMA*, which disables exponential smoothing and uses raw rtt_t, bw_t, loss_t for detection; and (ii) *No-Freeze*, which detects handovers but disables the window floor enforcement and notification masking. The event counter and error threshold are integral to the detection logic and cannot be removed without invalidating the algorithm; they are therefore not ablated.

Table I reports the average bandwidth and packet loss rate across all runs. Disabling EWMA reduces bandwidth by 14.25% (from the full AHD configuration of 12.00 Mbps to 10.29 Mbps) and increases the loss rate from 4.42% to 4.63%, confirming that smoothing is essential for filtering measurement noise and avoiding false positive detections. Disabling the freeze protection mechanism produces a substantially larger degradation: bandwidth drops by 21.25% (from the full AHD configuration of 12.00 Mbps to 9.45 Mbps) and the loss rate rises from 4.42% to 4.91%, approaching the performance of standard BBR. This demonstrates that the freeze protection mechanism is the primary contributor to AHD's performance improvement, while EWMA provides a necessary and complementary filtering contribution.

TABLE I
ABLATION STUDY RESULTS

Variant	**Bandwidth** (Mbps)	**Packet Loss Rate** (%)
EWMA + Freeze	12.00	4.42
No EWMA	10.29	4.63
No Freeze	9.45	4.91

V. CONCLUSION AND FUTURE WORK

This paper has proposed and evaluated AHD, an adaptive handover detection algorithm designed to address the transmission performance degradation caused by frequent handovers in LEO satellite networks. Unlike existing approaches, AHD does not rely on proprietary hardware feedback or fixed handover schedules, operating solely on observable end-to-end metrics. Integrated as a wrapper around BBRv3 within the QUIC protocol and evaluated in Mininet-based simulations driven by real Starlink traces, AHD identifies network state transitions caused by handover and suppresses the unnecessary *cwnd* reductions that afflict traditional CC algorithms. The experimental results demonstrate that AHD achieves up to 32% throughput gain and 11% packet loss reduction over standard BBR, while also outperforming StarQUIC, offering a solution for satellite Internet protocol optimization.

Despite these promising results, several directions remain open for future work. First, the current detection mechanism relies on a limited set of transport-layer observables. Incorporating richer signals, such as packet reordering patterns, jitter characteristics, or congestion window dynamics, could improve detection accuracy in complex orbital environments. Second, future work will include a quantitative comparison with SaTCP and machine learning-based handover detection, as well as evaluation on ISL-enabled topologies and multiple PoPs, once suitable emulators or labeled datasets become available. Third, all experiments were conducted in a Mininet-based simulator using historical Starlink traces. Field deployment tests on live LEO satellite networks such as Starlink and OneWeb are needed to validate the practical applicability of AHD. Fourth, this paper focuses on single-flow performance. The fairness of AHD when coexisting with multiple competing flows, including other AHD flows and flows governed by legacy CC algorithms, has not been studied. In principle, AHD's freeze mechanism only temporarily suppresses *cwnd* reduction during handover transients and otherwise follows BBRv3's rate control, which is known to maintain fairness in static environments; however, rigorous multi-flow evaluation is left as future work.

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